Project Name: Event Ticket & Queue Management System

Project Summary:

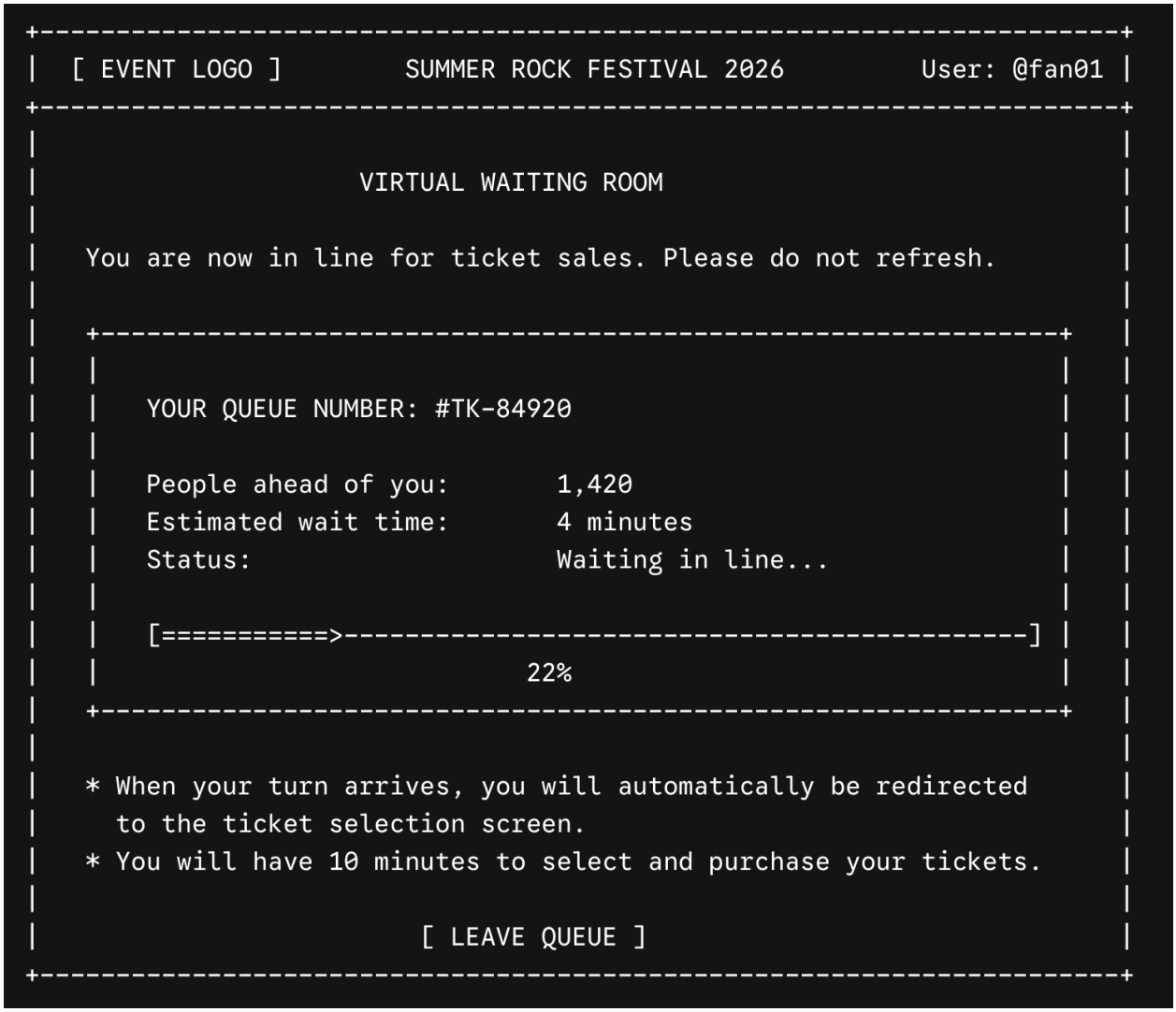
A high-throughput ticketing platform designed to manage extreme traffic spikes during popular concert or event sales. It implements a token-based virtual waiting room to buffer incoming requests, serialize checkout traffic, and eliminate race conditions or ticket overselling using atomic inventory holds and automated reservation timeouts.

Core Architectural Alignment:

- **Pipeline Architecture (Slide 8.1):** Unidirectional order processing using distinct filter types—Producer (queue intake), Tester (availability/hold verification), Transformer (pricing/discounts), and Consumer (order persistence to database).
- **Microkernel Architecture (Slide 8.2):** A stable core booking engine that integrates plug-in components via standard contracts for different payment methods (e.g., Stripe, PromptPay) and promotion/pricing rules.

Scope for Requirement Model Deliverables :

- **Primary Actors:** Customer and Organizer/Admin.
- **Core Use Cases (5–7 total):**
  - Join Waiting Room
  - Select & Hold Tickets
  - Checkout & Pay
  - View Digital Ticket
  - Create / Manage Event & Tiers
  - Monitor Live Queue & Sales
- **Primary Workflow (for Activity Diagram & SSD):**
  - *Flow:* Customer enters queue → Receives wait position → Admitted to select ticket tier → System places temporary hold (10-minute TTL) → Customer completes payment → System issues verifiable digital ticket.
- **Domain Classes (Entities):**
  - `Event`, `TicketTier`, `QueueToken`, `Reservation`, `Payment`, `DigitalTicket`.
- **UI Mockup Target:**
  - The Customer Queue screen (showing real-time queue position and progress)



***EX**

Member 3 Project II: Behavioral Models & UI

- Draw the **System Sequence Diagram (SSD)** for the core booking and payment flow.
- Draw the **Activity Diagram (Workflow)** for entering the queue, holding a ticket, and checkout.
- Create mockups/wireframes for the **User Interface (UI)** (Queue screen, Ticket selection/countdown, Confirmation).

Member 4 Lab 9 Task 1: Bridge Pattern
Member 5 Lab 9 Task 2: Mediator Pattern
Member 6 Lab 9 Task 3: Proxy Pattern